Lecture – 8
Advanced Sorting Methods –Part 3

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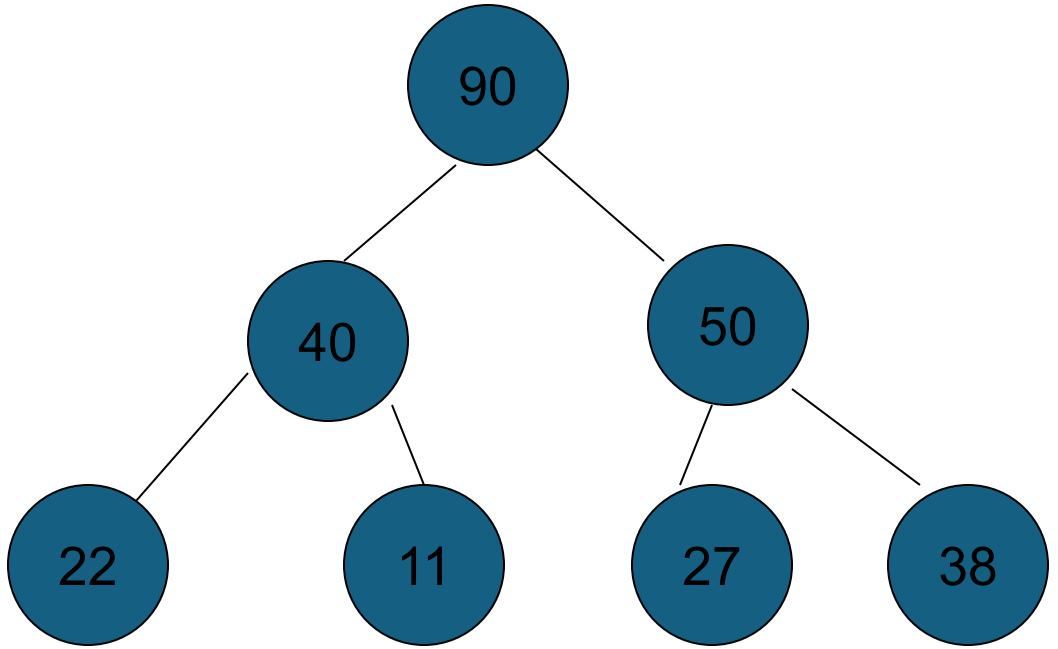
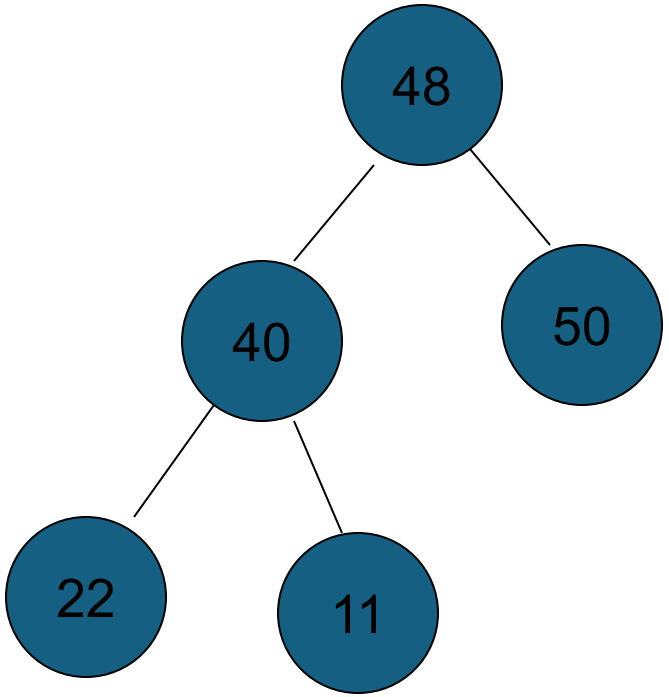
Heaps, Heap sorts and Priority Queues

- Heap
- A heap of size n is a B-tree of n nodes such that the values of each node is >= the value of its children. (if any)

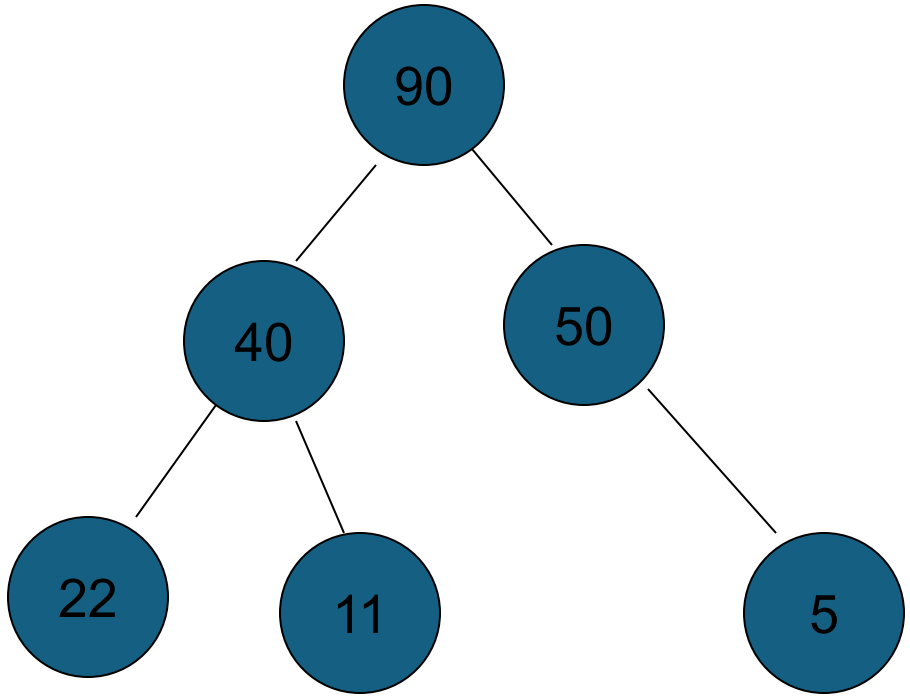
Examples



A Heap



Not a heap [ie 50 <48]



Not a Heap [Gaps]

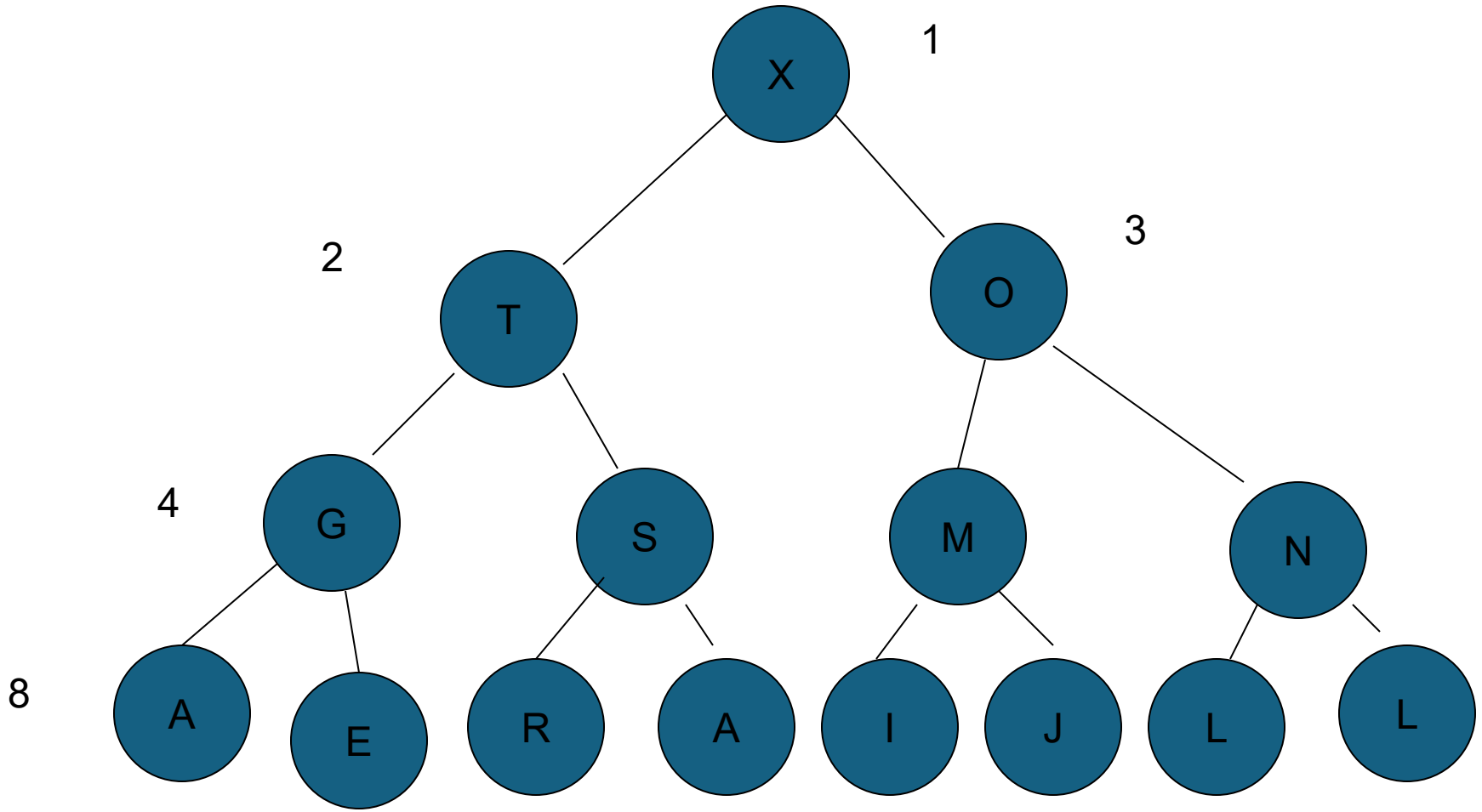
Properties

- A heap, however, is a special kind of binary tree that leaves no gaps in an array implementation.
- The definition of the heap data type is as follows.

A heap is a binary tree satisfying the following restrictions.

- All leaves are on adjacent levels.
- All leaves on the lowest level occur at the left of the tree
- All levels above the lowest are completely filled.
- Both children of any node are again heaps.
- The values stored at any node is at least as large as the values in its two children.

Array representation of a heap

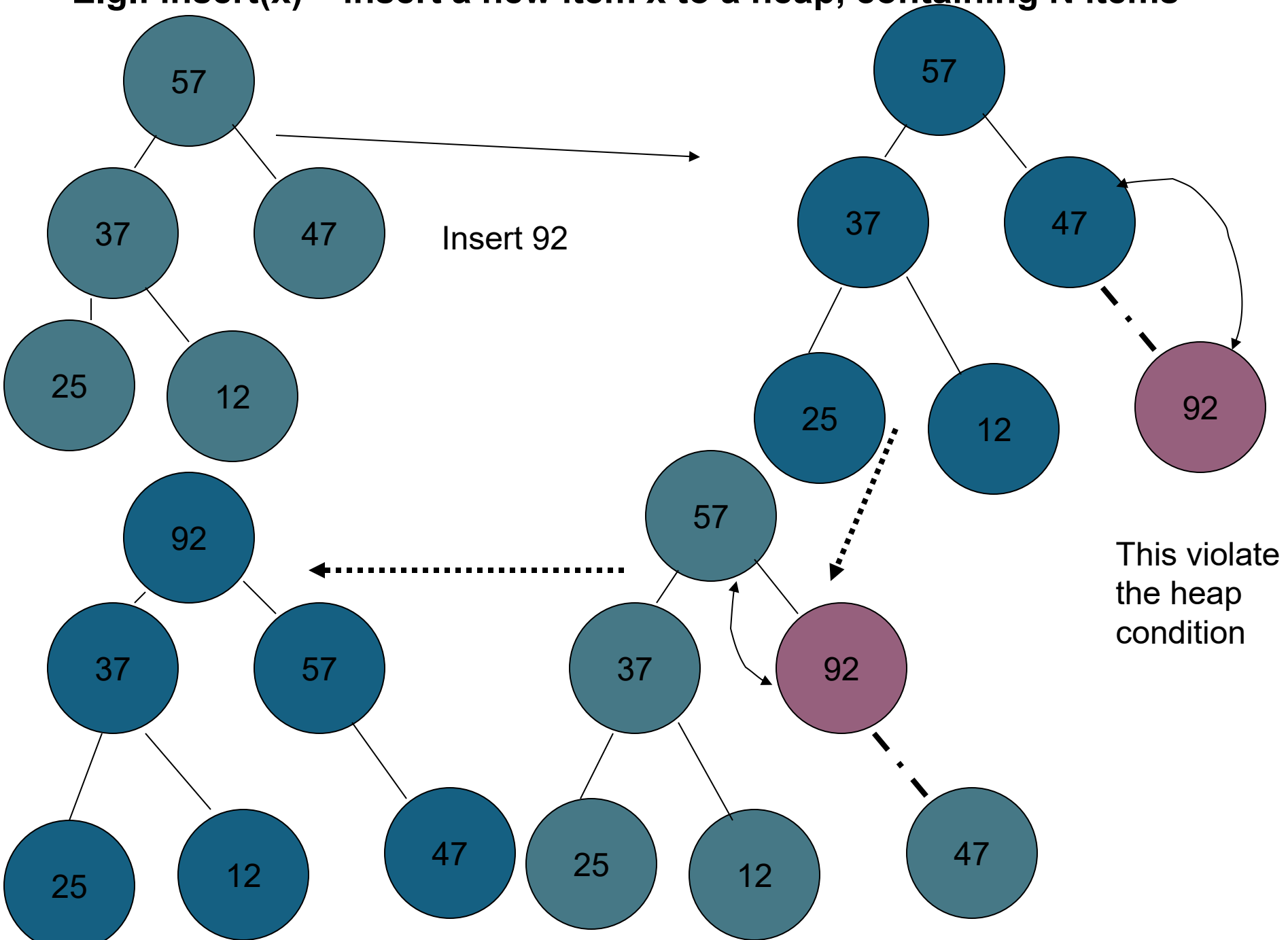


The tree may be easily constructed from the array by noticing that the left child of the k as 2k and right child 2k+1.

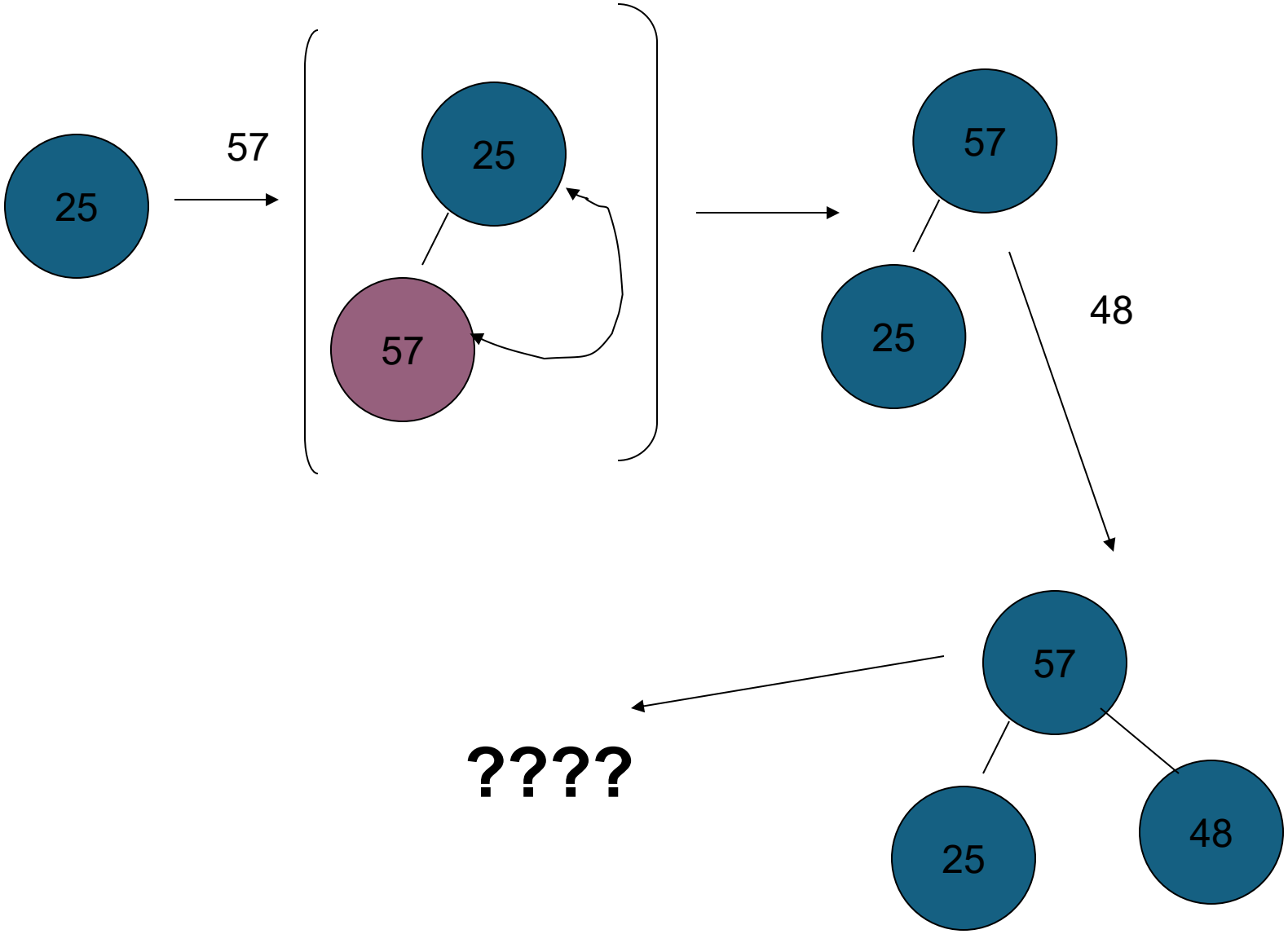
Algorithms on Heaps

- The algorithms on heap works by first making a simple structural modification which could violate the heap condition, the traveling through the heap modifying it to ensure that the heap condition is satisfied everywhere.
- Some of the algorithm travel through the heap from bottom to top, others top to bottom.

E.g.: insert(x) – insert a new item x to a heap, containing N items



Creating a Heap of size 8. The original file is 25,57,48,37,12,92,86,33

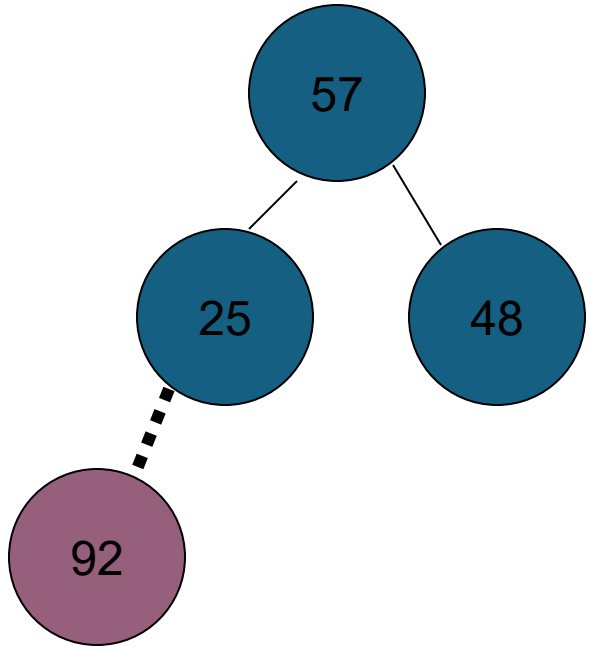


- We want to build and maintain a data structure containing records with numerical keys (priorities) and supporting some of the following operations.
- Construct a priority queue from given N items
- Insert a new item (with a priority)
- Replace the largest item with a new item.(unless new item is larger)
- Change the priority of an item.

- To be able build a heap, it is necessary, first implement the insert operation. This operation will increase the size of the heap by one. N must be incremented. Then the record to be inserted is put into A[N] , but this may violate the heap property (ie. New node is grater than its parent) then the violation can be fixed by exchanging new node with its parent .

- The code for this method is straight forward.
- Insert : add a new item to a[n], then upheap[n] fix the heap condition violation at N.

Pseudo code for insertion



Insert (X)

N++

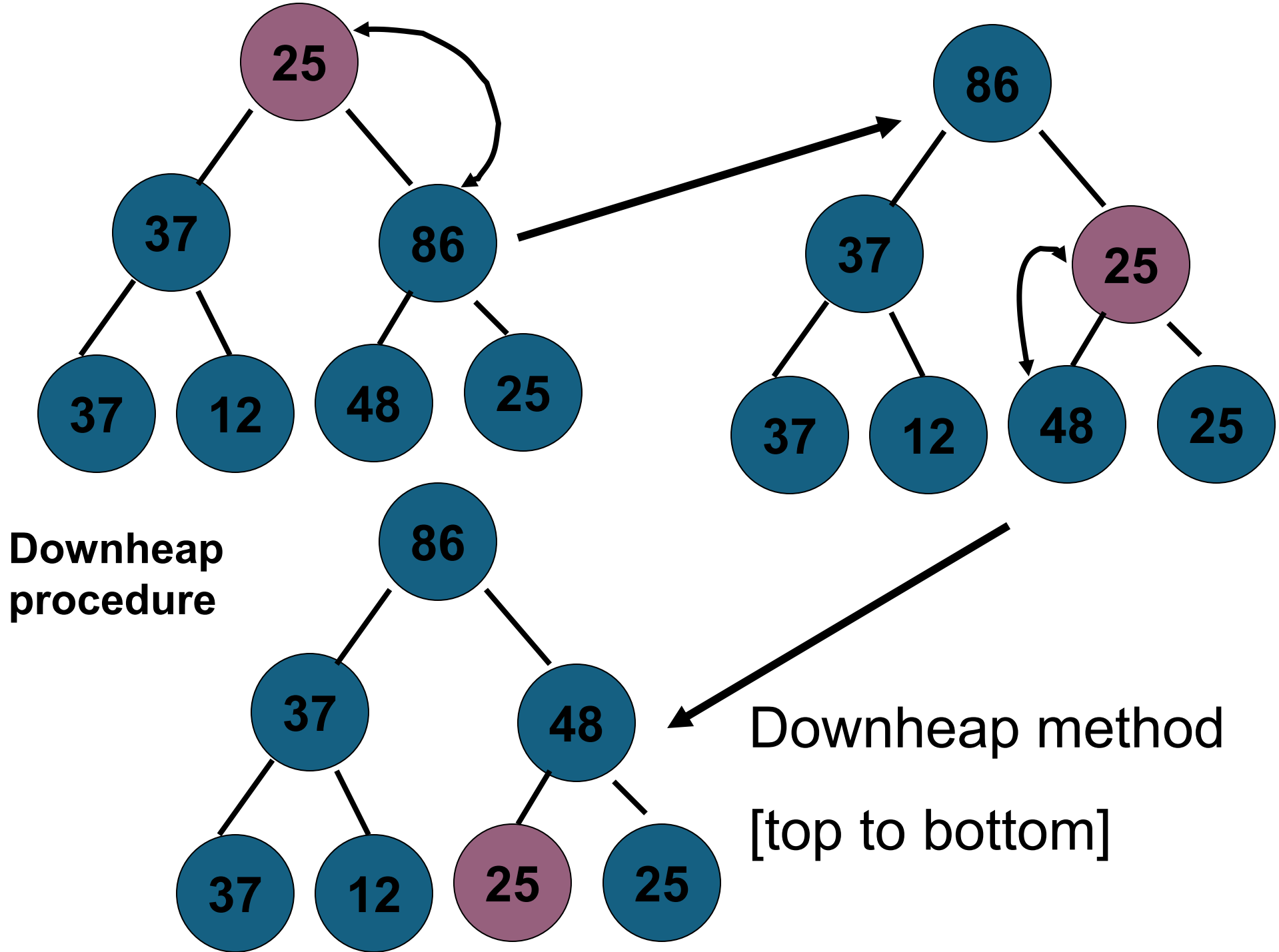
A[N]=x

Upheap{N]

Upheap procedure ??

Downheap Method

- If the element at node k is smaller than its children, that element is moved down the heap without violating the heap property at any nodes touched.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Downheap
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Downheap method
[top to bottom]

Remove the largest item-pseudo code

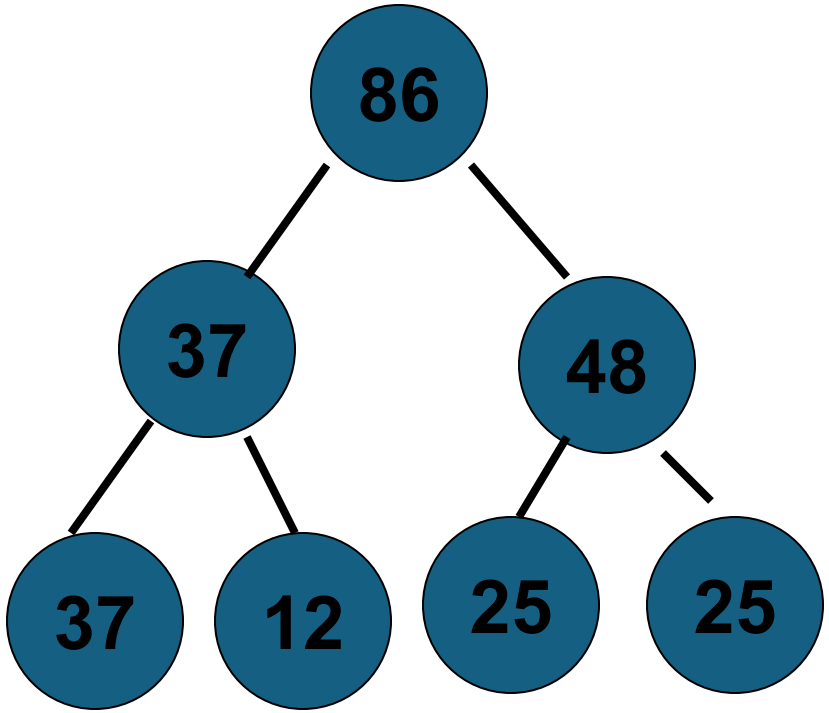
Remove the highest priority element ( ) ;

return = A[0];

A[0] = A[N-1];

N--;

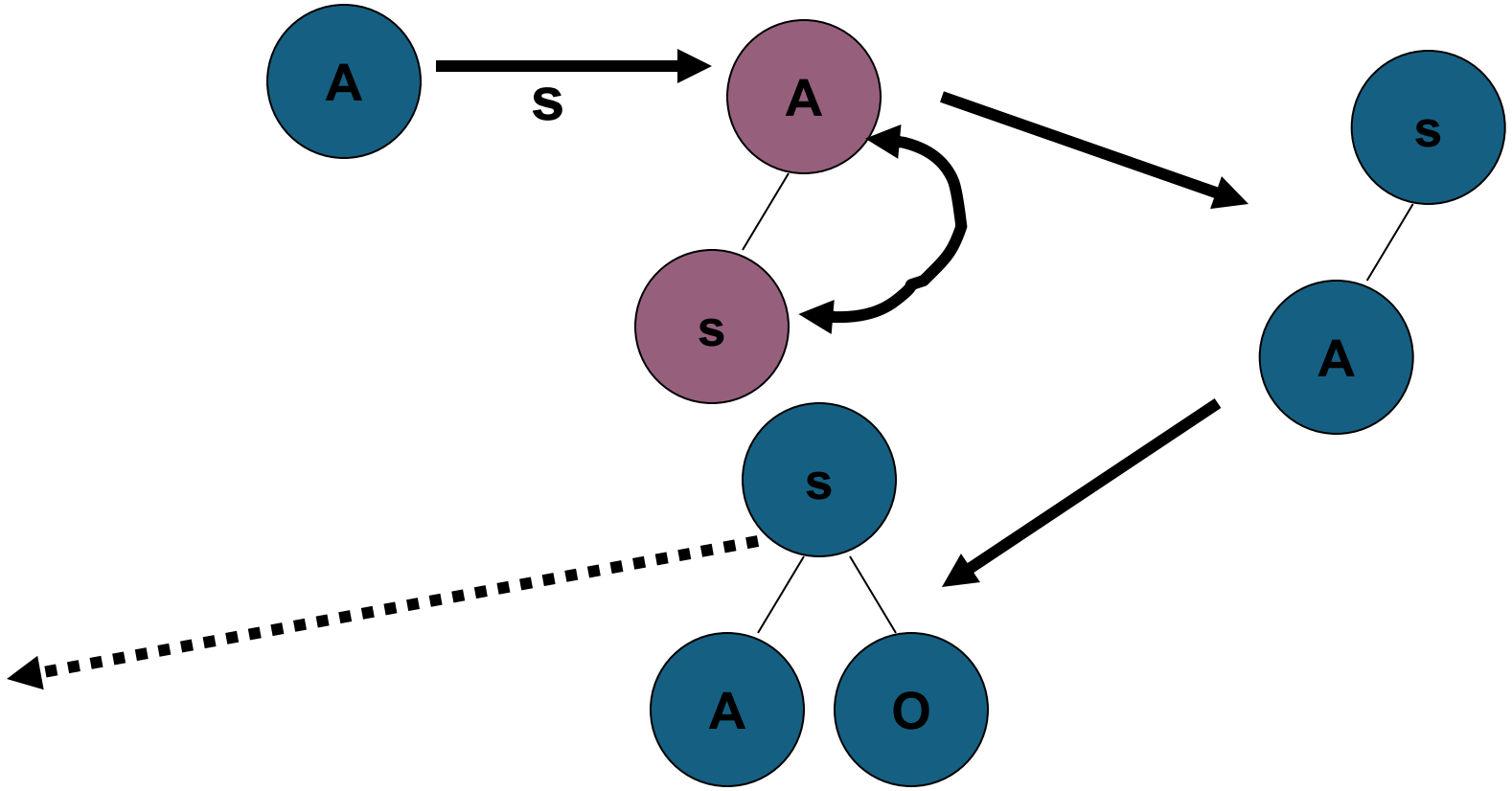
Upheap[N];



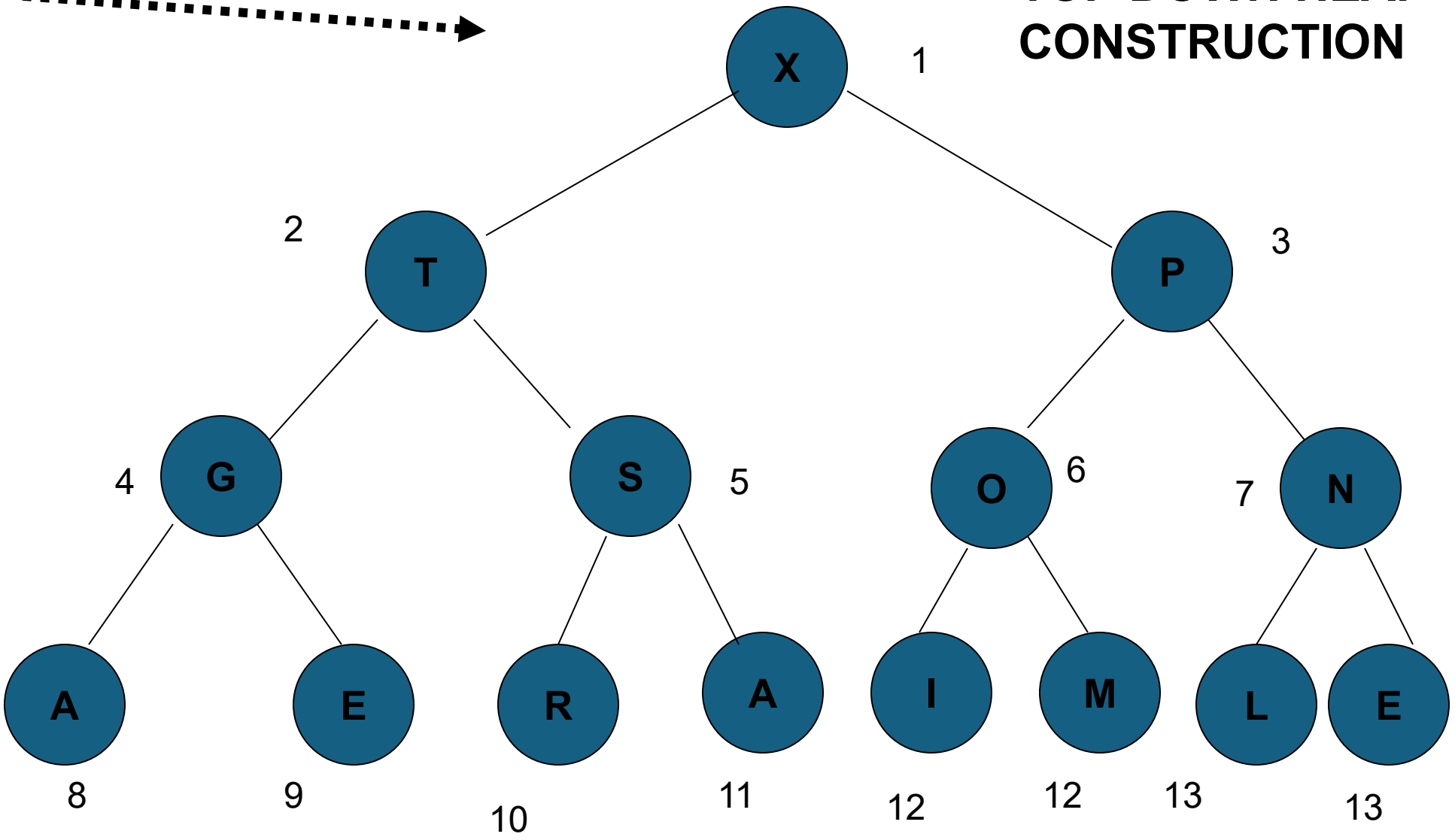
The return value is set from A[0] and then the element from A[N-1] is put into A[0] and the size of the heap is decremented, call down heap procedure to fix up the heap condition everywhere.

Heap Sort

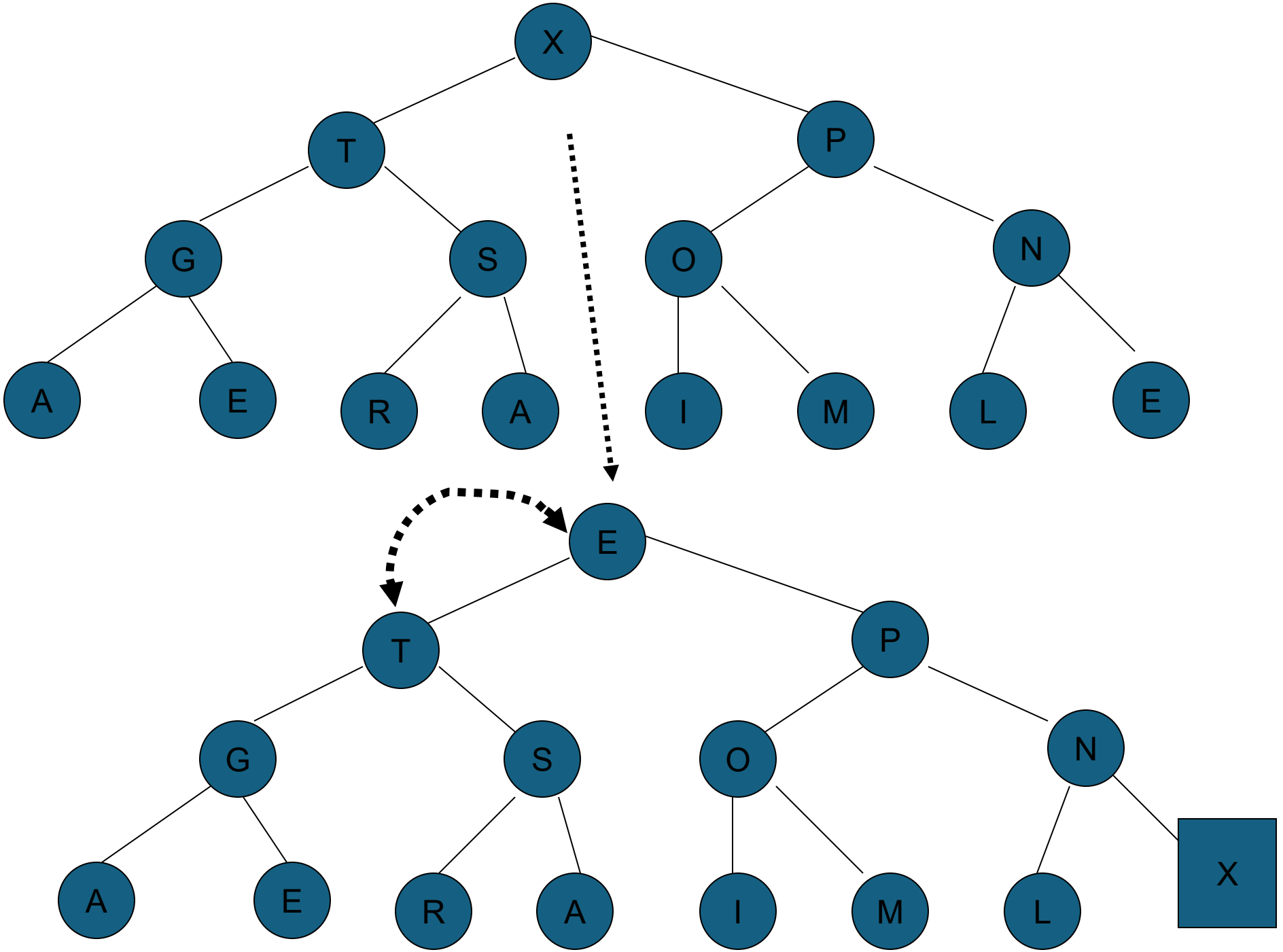
Eg : A SORTING EXAMPLE



TOP DOWN HEAP CONSTRUCTION



Above figure shows the heaps constructed as the keys A SORTING EXAMPLE are inserted in that order into an initially empty heap.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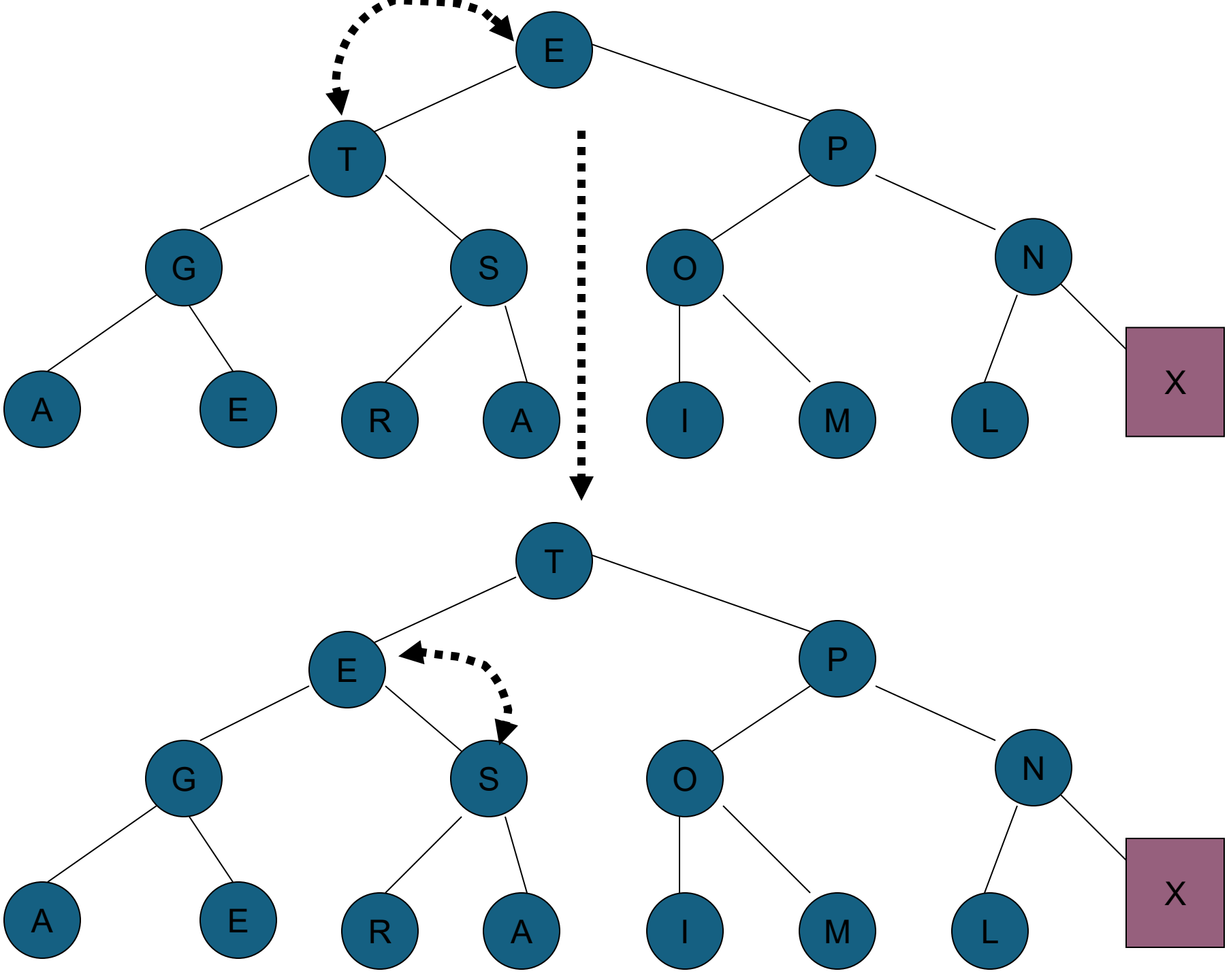
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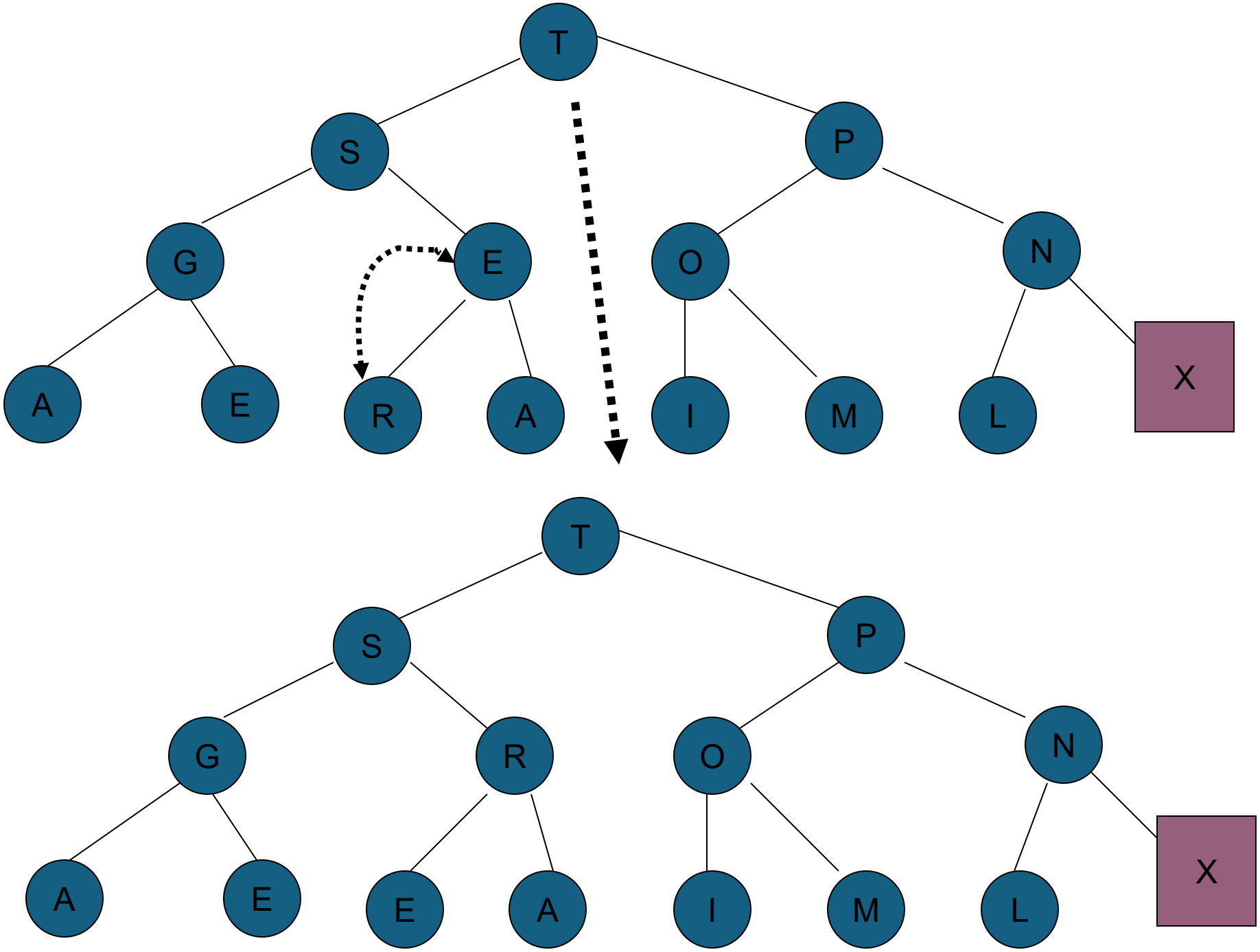
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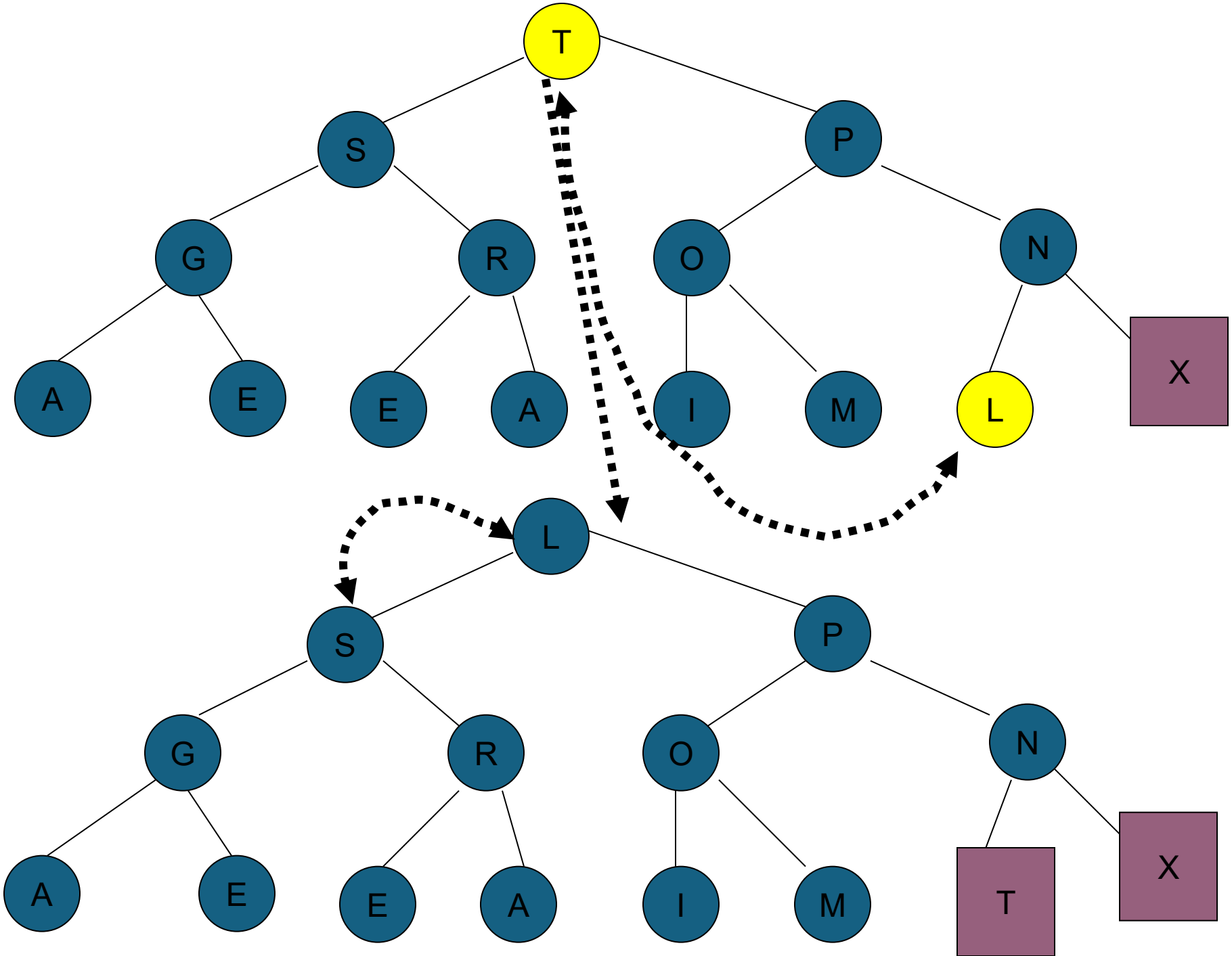
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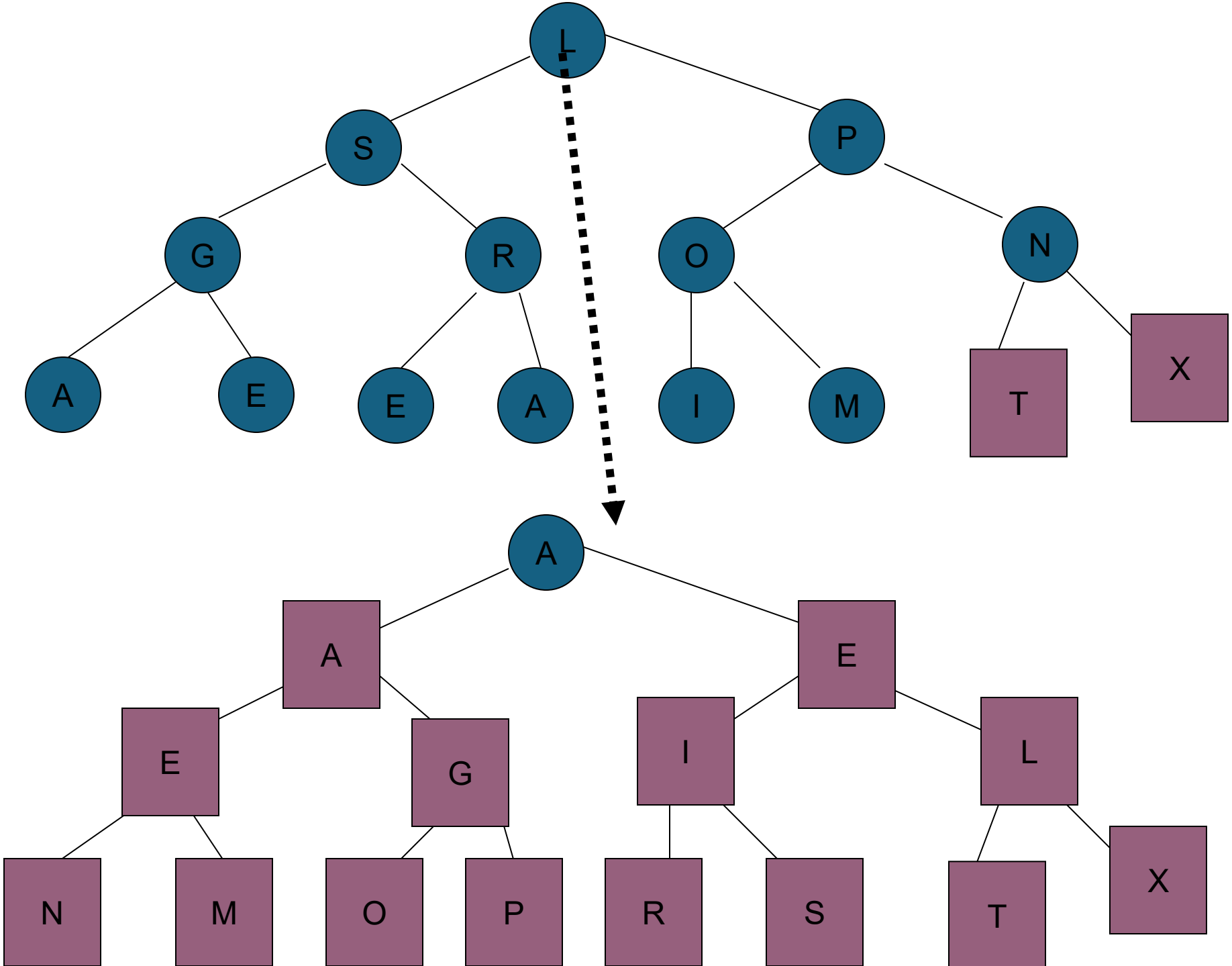
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- 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
- A[I] A A E E G I L N M O P R S T X

- We were already noted that remove can be implemented by exchanging the first and last element and decrementing n by one and calling downheap from one.